
بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

SPECTRAL REFLECTANCE

- **The average amount of incident radiation reflected by a surface at some wavelength interval is called '*Spectral reflectance* ρ_λ" and normally expressed as a percentage**

$$\rho_\lambda = E_R(\lambda) / E_I(\lambda)$$

$$= \frac{\text{Energy of wavelength } \lambda \text{ reflected from the object}}{\text{Energy of wavelength } \lambda \text{ incident upon the object}} \times 100$$

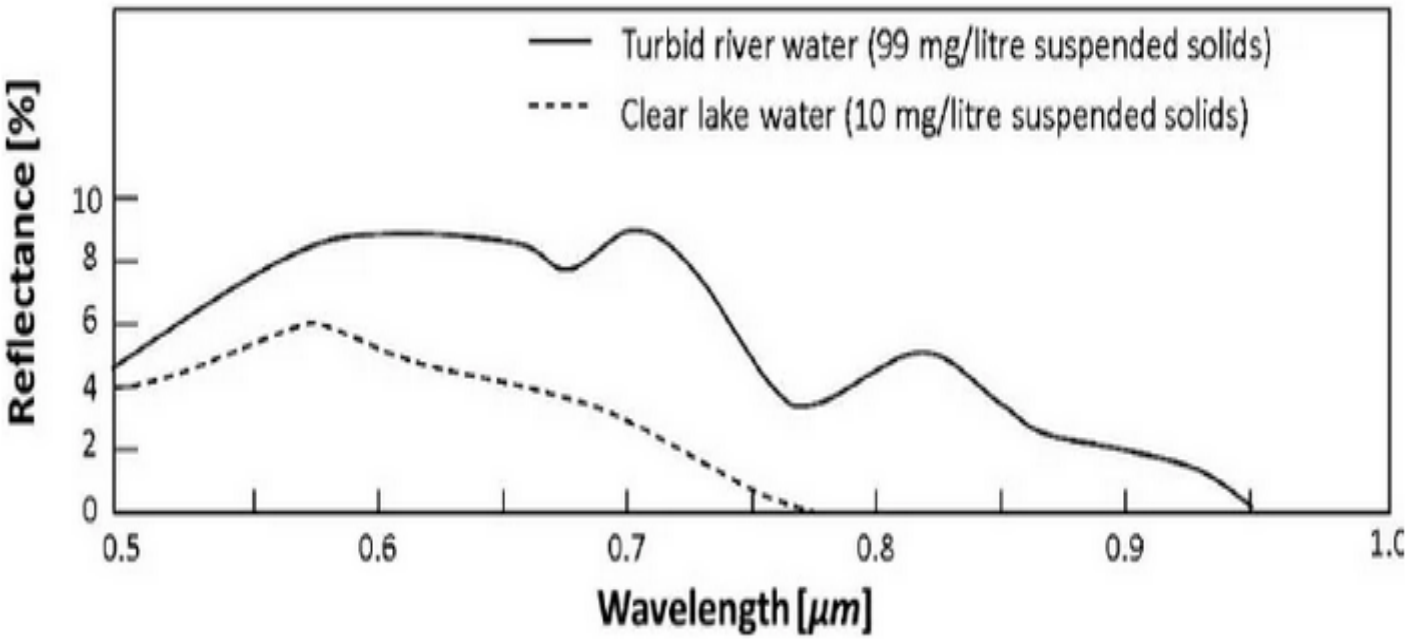
SPECTRAL REFLECTANCE CURVE /PROFILE

- **Graph of spectral reflectance of an object as a function of wavelength is termed as "Spectral reflectance curve/profile"**
- **Reflectance curves are made for the optical part of the electromagnetic spectrum (up to 2.5-3 μm)**
- **Once the incident radiations interacts with a specific surface, and how this surface behave is described as " SPECTRAL RESPONSE" of that surface**

INTERACTION OF EMR WITH WATER

- **Possible reflectance from Water bodies**
  - **Surface reflectance**
  - **Volume reflectance**
  - **Bottom reflectance**
- **Reflectance of Water is affected by**
  - **WLs**
  - **Depth**
  - **Impurities**
  - **Surface condition**

Spectral Reflectance of Water

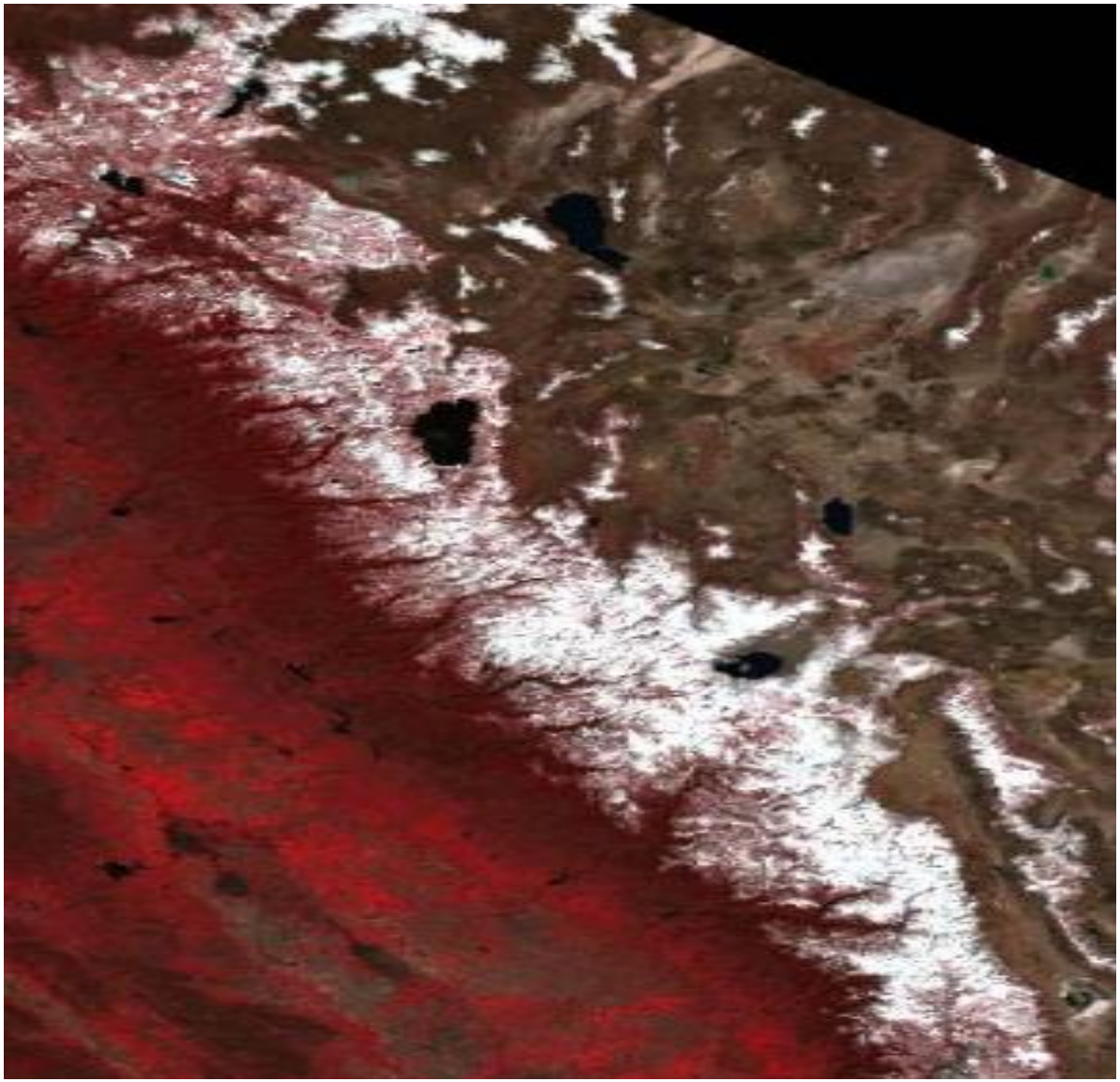


SPECTRAL REFLECTANCE OF SNOW

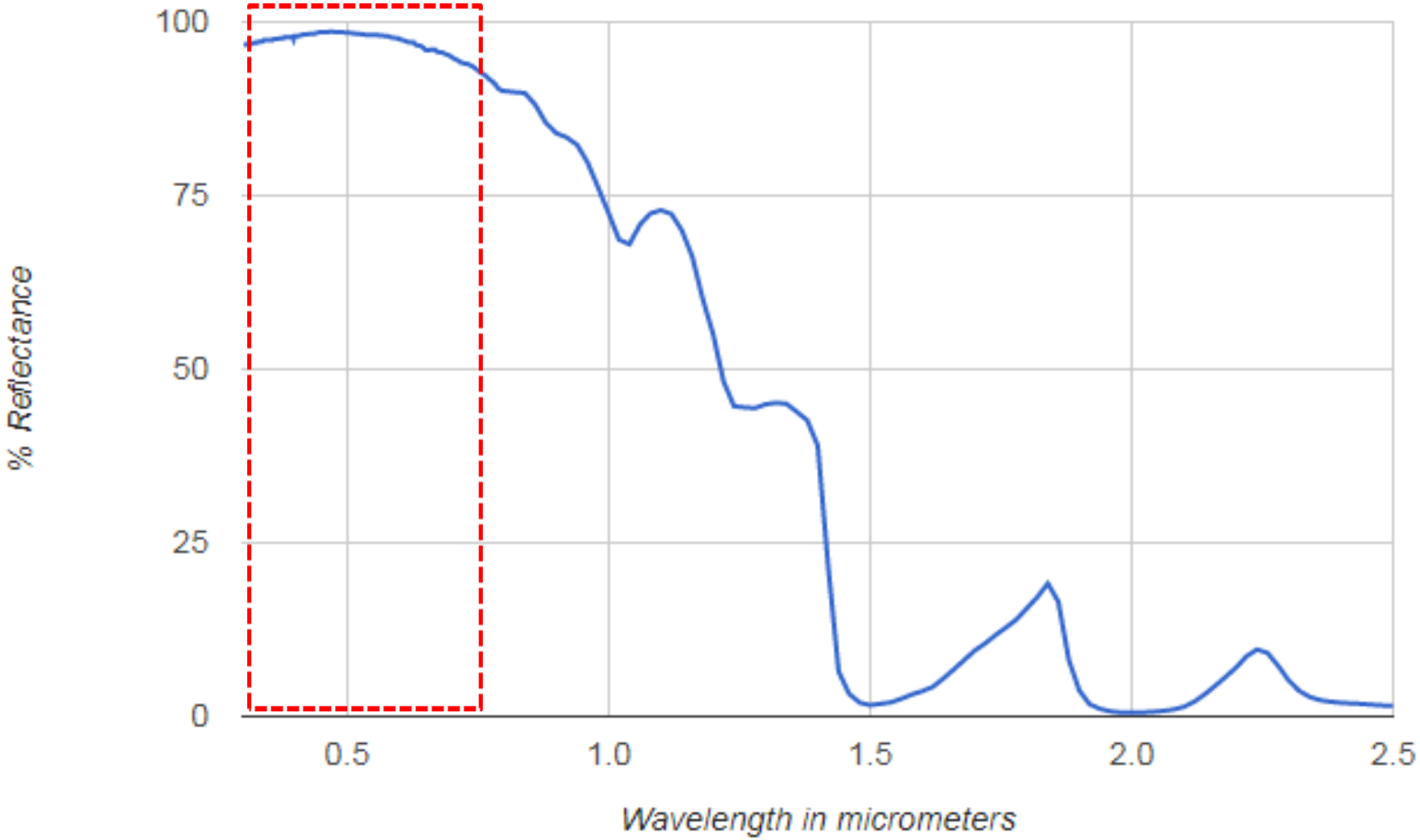
Spectral Reflectance- Snow

- **The reflectance of the snowpack depend on the**
  - **Wavelength**
  - **Age (time)**
  - **Grain size distribution of the snow**
  - **Depth and density of the snowpack**
  - **Size and amount of the impurities**

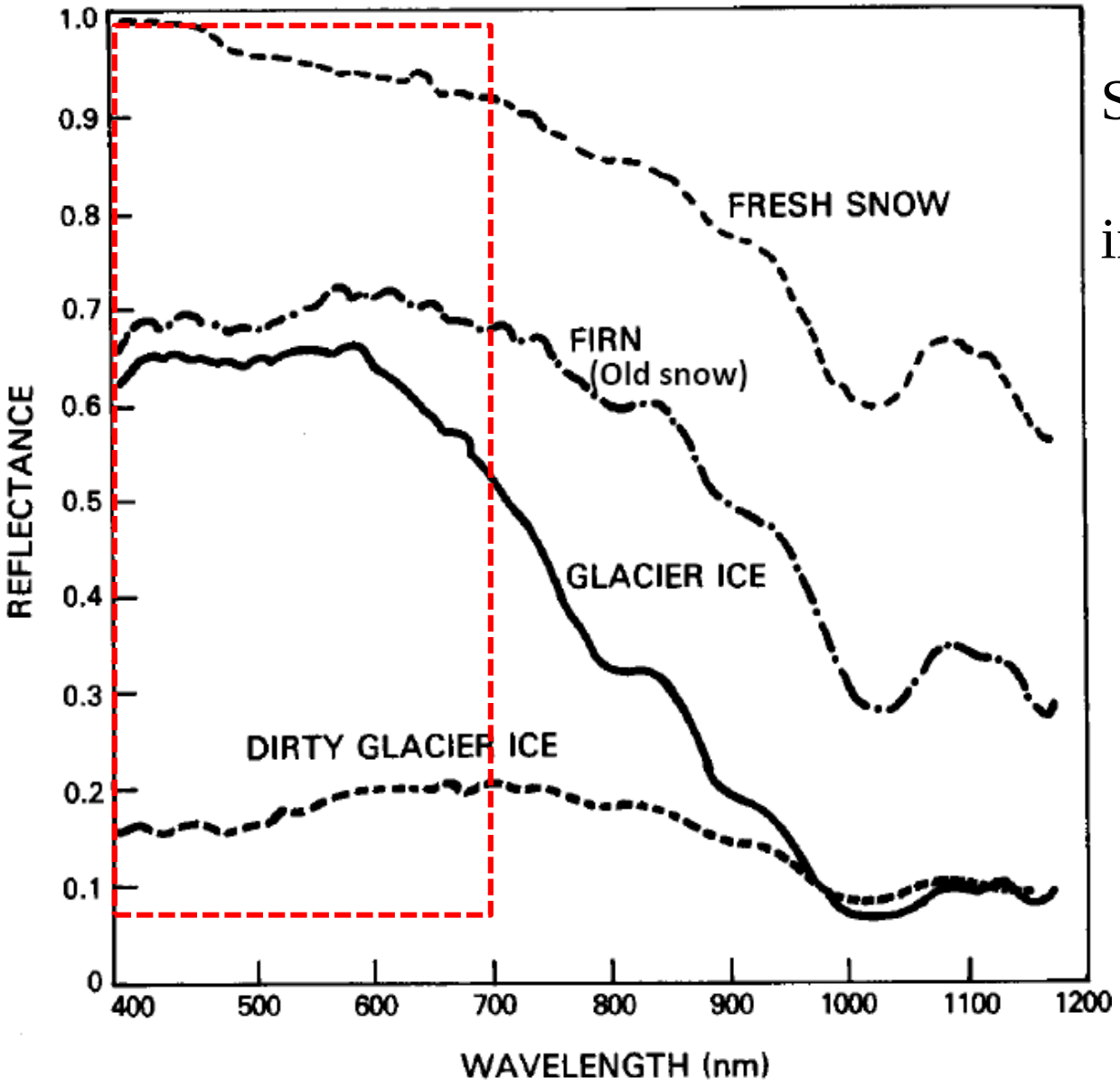
Spectral Reflectance- Snow



Spectral Reflectance of Snow

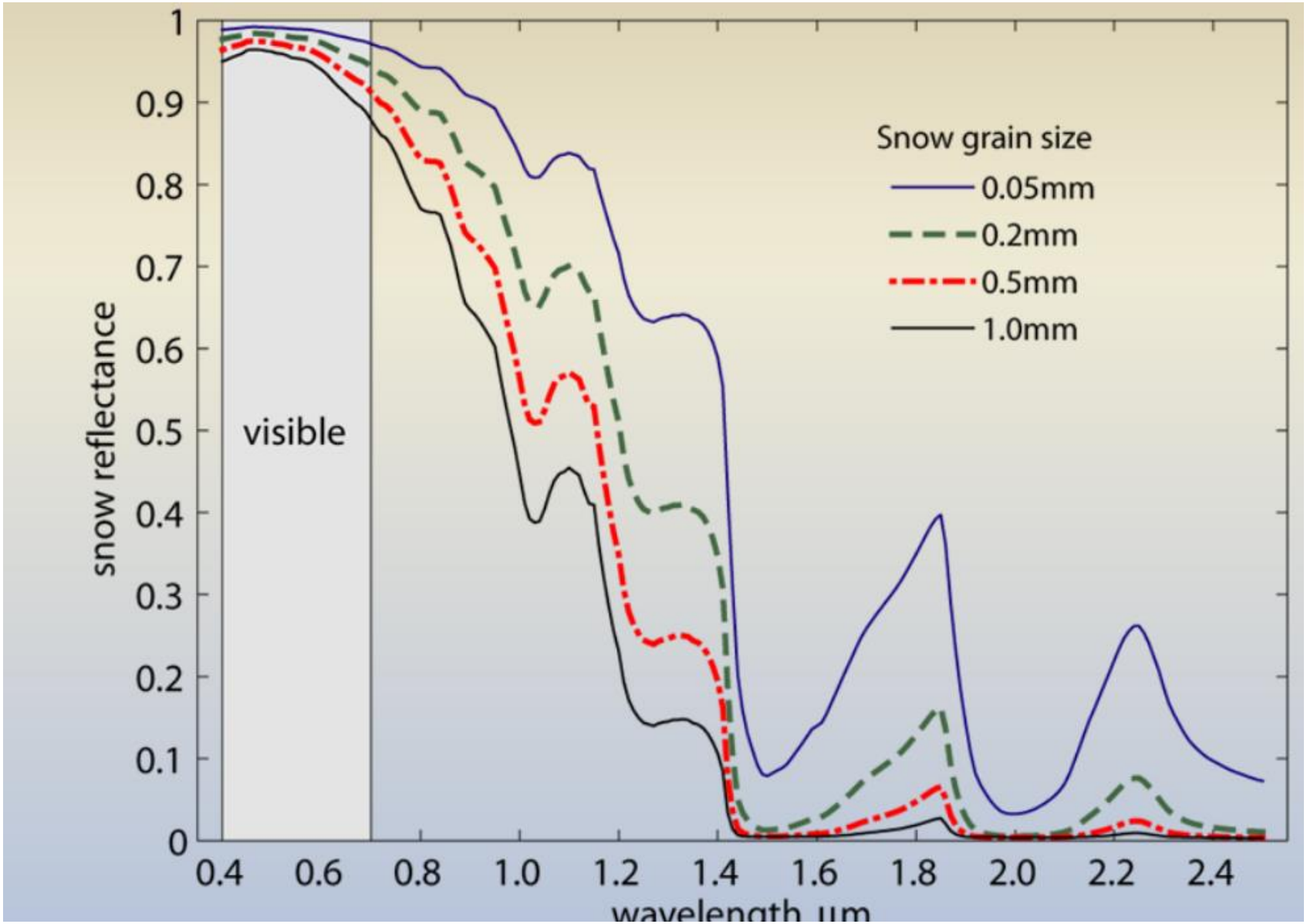


Spectral Reflectance- Age/ Impurities

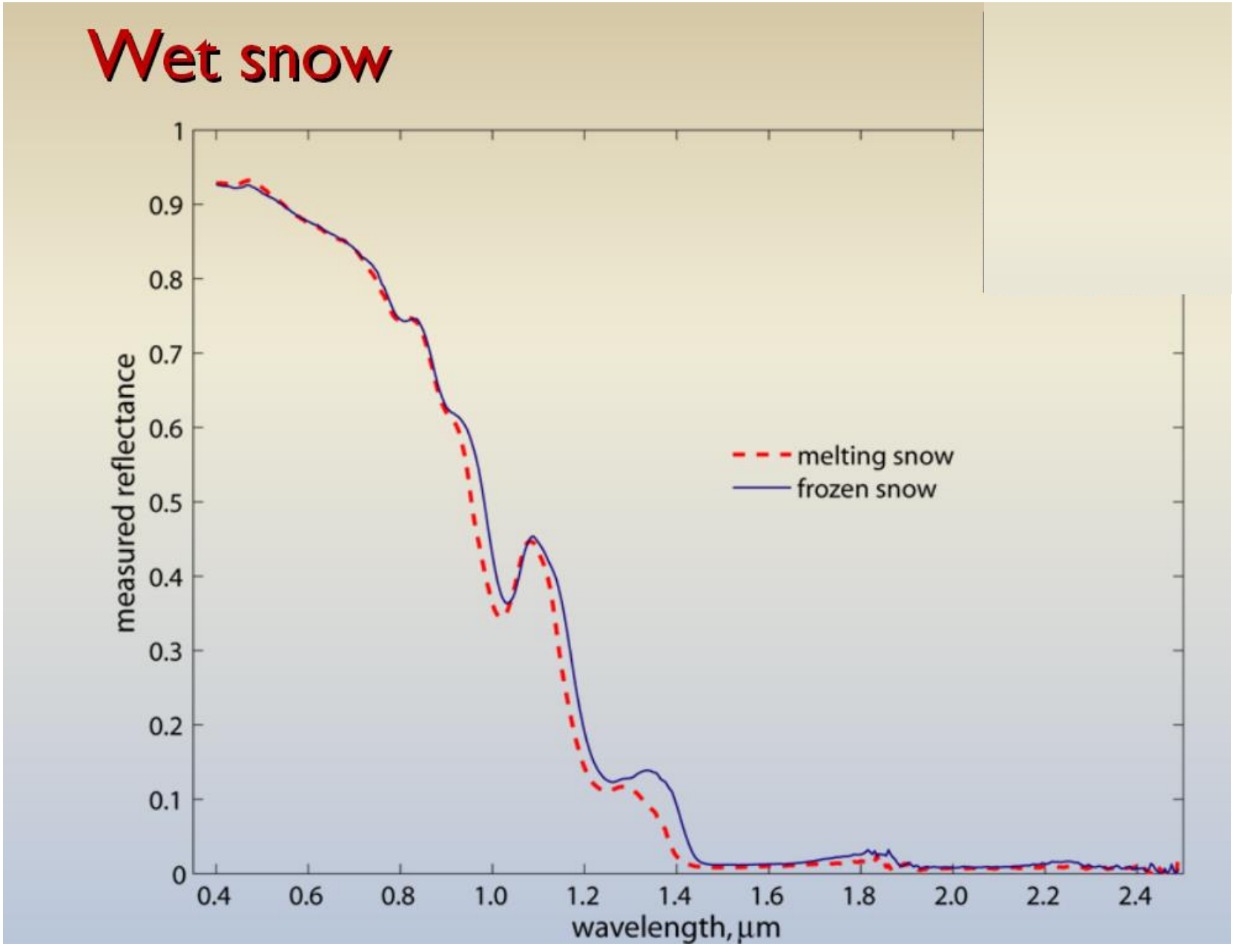


Size and amount of the impurities

Spectral Reflectance of Clean Snow- Grain Size



Melting Vs Frozen Snow Snow



SPECTRAL REFLECTANCE

ROADs and Impervious Surfaces

Spectral Reflectance- Impervious Surfaces

- **The reflectance depend on the**
  - **Wavelength**
  - **Age**
  - **Type of the material**
  - **Surface Condition**

Impervious Surfaces



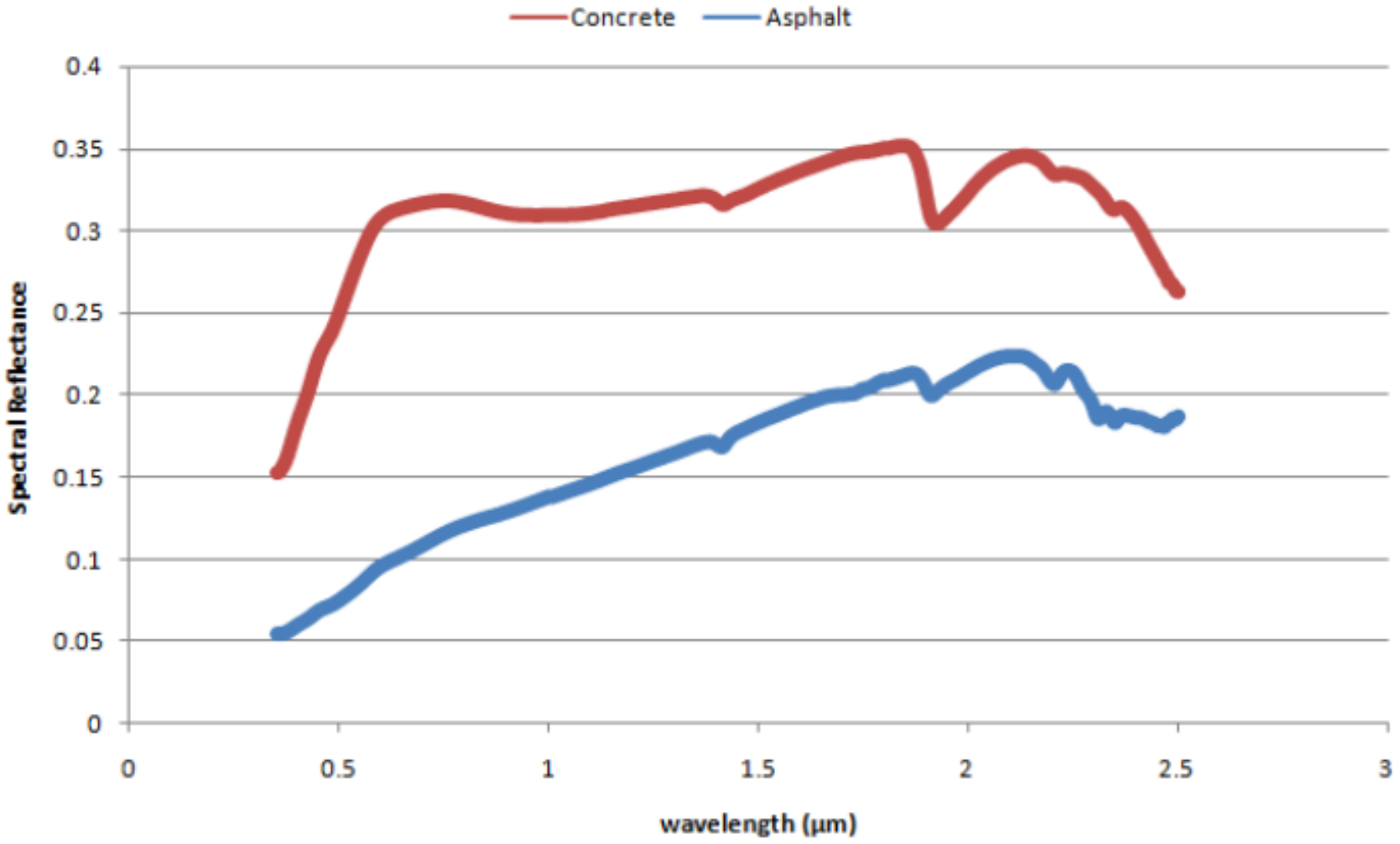




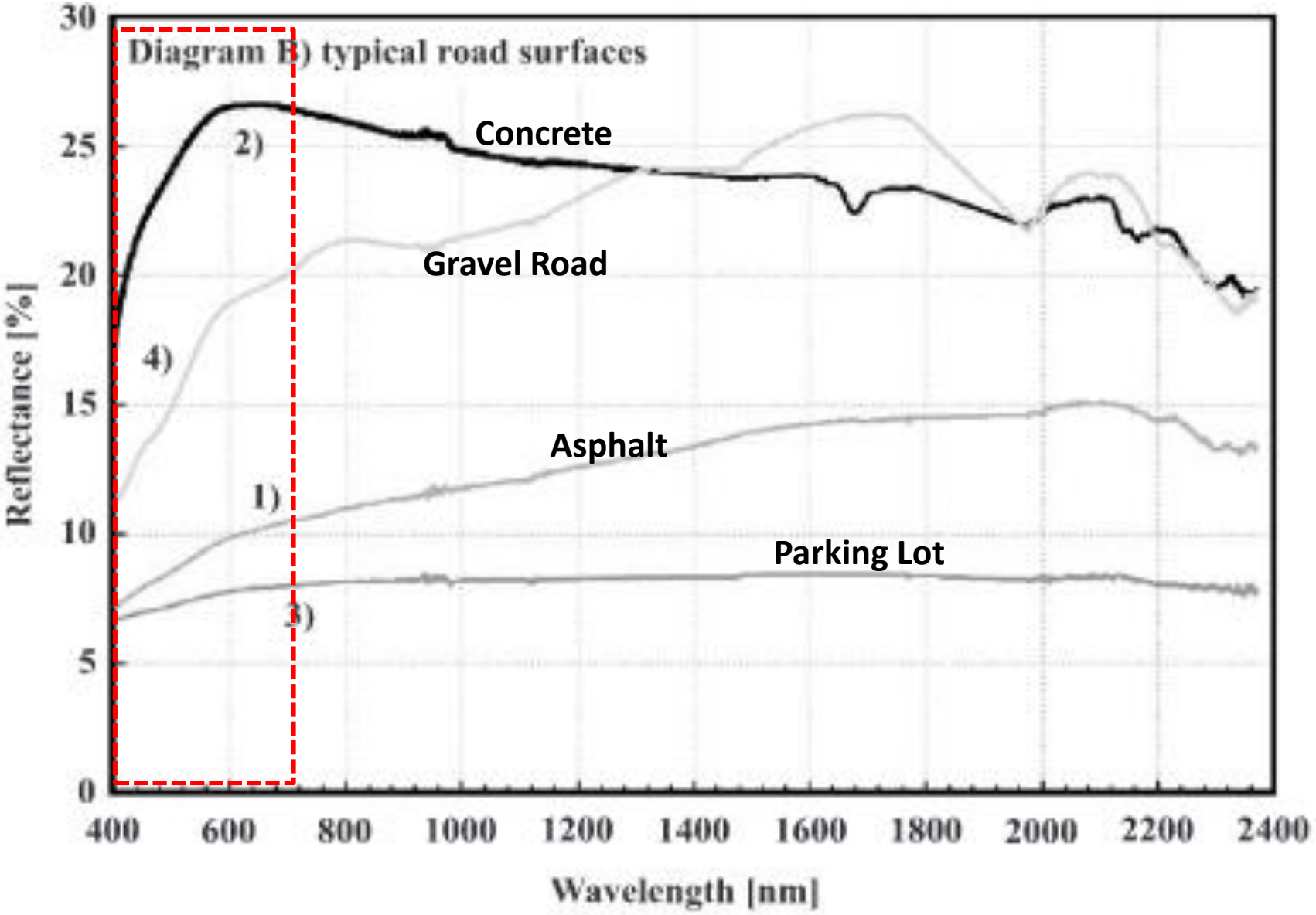




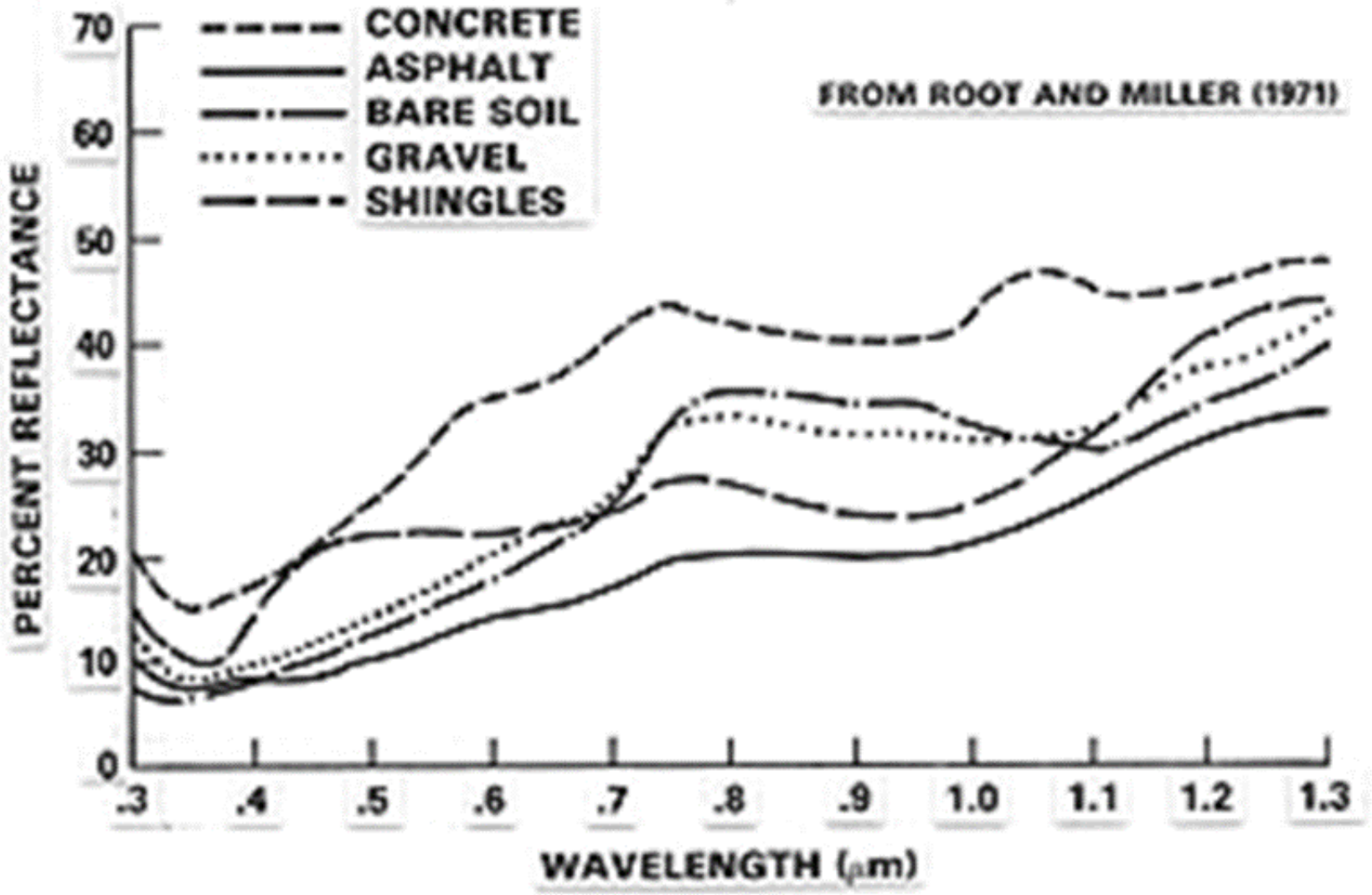
Spectral Reflectance- Impervious Surfaces

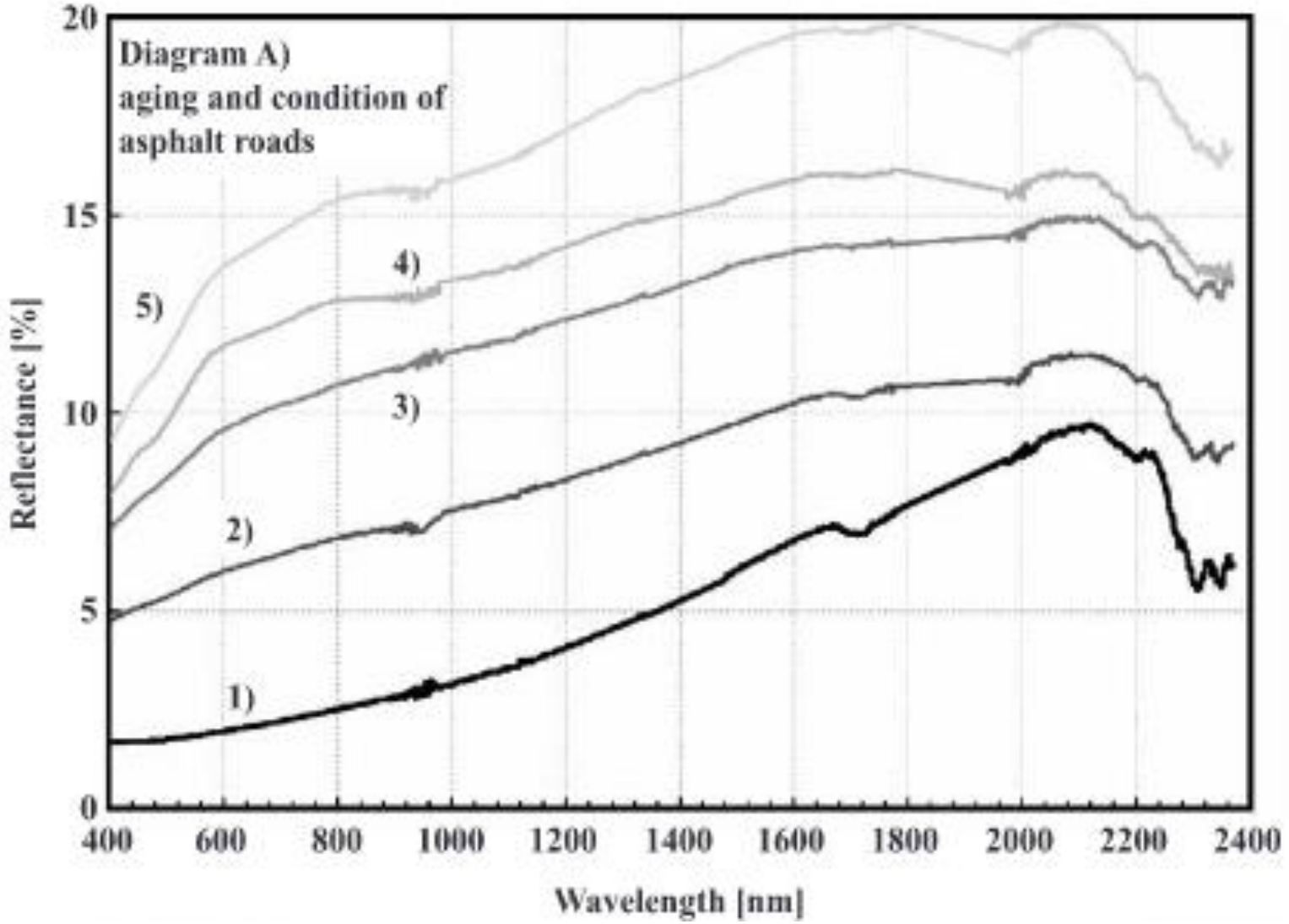


Spectral Reflectance- Road and Parking Surfaces

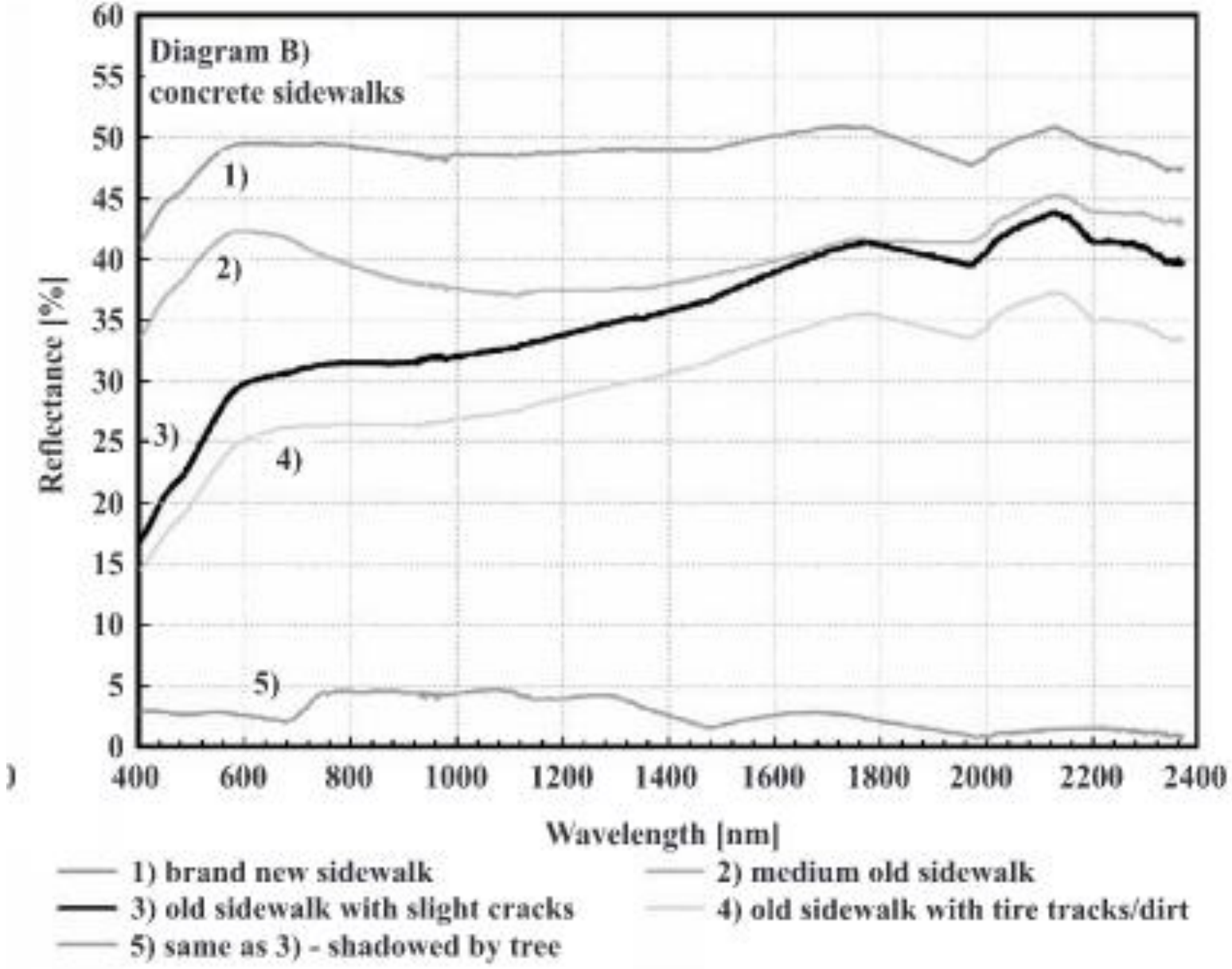


Spectral Reflectance- Different Materials

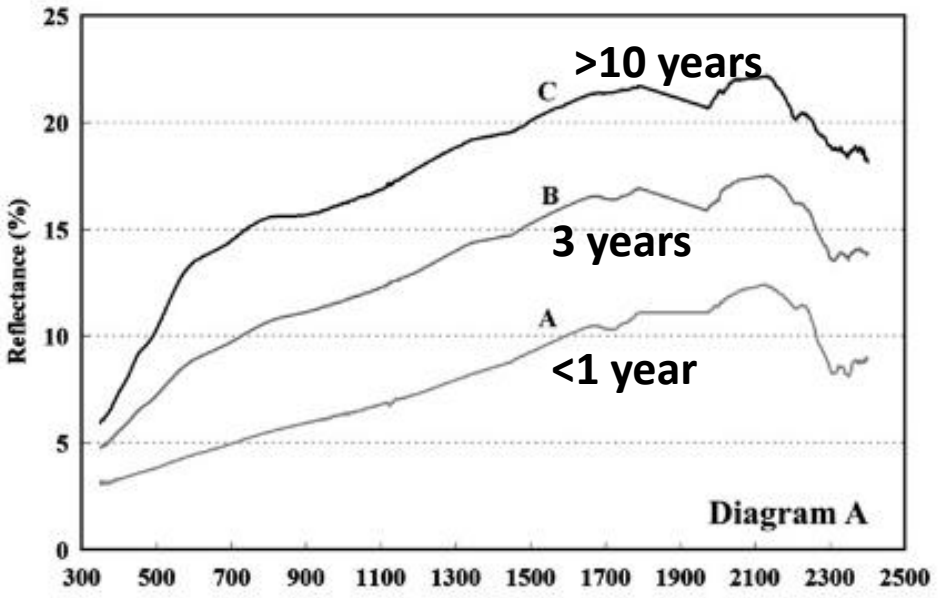


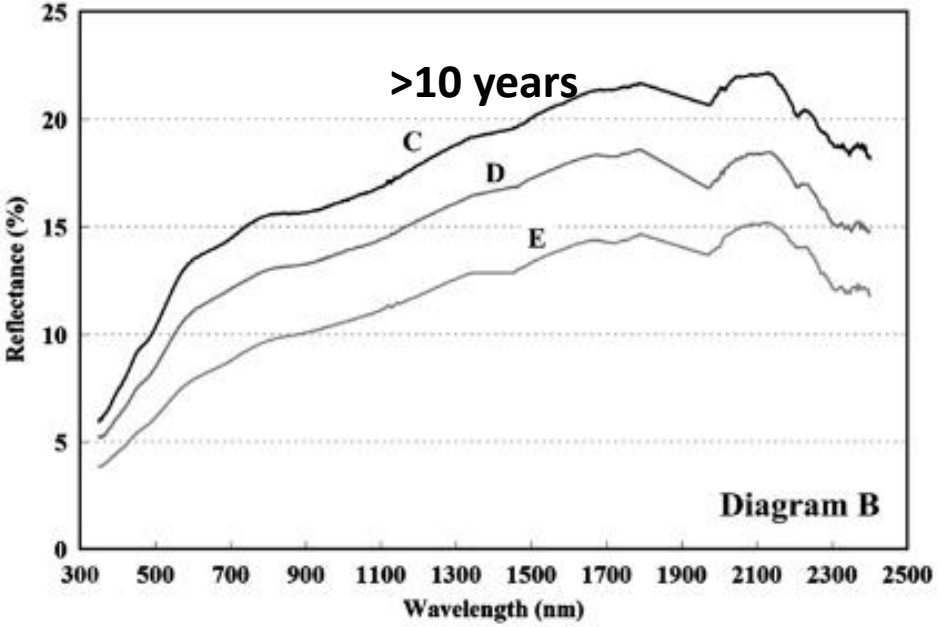


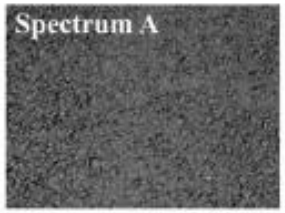
1. **Fresh Asphalt Road (Constriction time)**
2. **New Asphalt Road**
3. **Refurbished Road**
4. **Old Asphalt Road (Fair Condition)**
5. **Old Asphalt Road (Very poor Condition)**


Diagram B)
concrete sidewalks
Reflectance [%]
Wavelength [nm]
1)
2)
3)
4)
5)
60
55
50
45
40
35
30
25
20
15
10
5
0
400
600
800
1000
1200
1400
1600
1800
2000
2200
2400
1) brand new sidewalk
2) medium old sidewalk
3) old sidewalk with slight cracks
4) old sidewalk with tire tracks/dirt
5) same as 3) - shadowed by tree

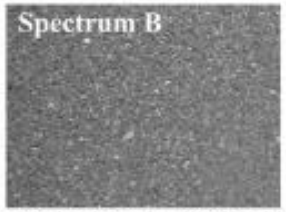
AGE



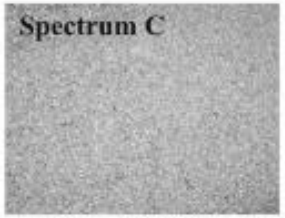




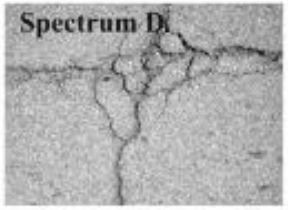
Age:<1yr/PCI:99/SI:100



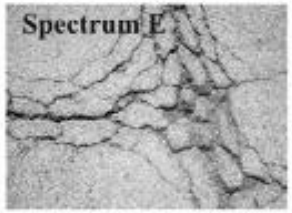
Age:3yrs/PCI:86/SI:100



Age:>10yrs/PCI:32/SI:63



Low severity crack



Severe alligator crack

